The empirical recurrence for OEIS A316172, recovered from its tail

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Abstract

OEIS A316172 records "Empirical recurrence of order 64" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 70$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 67$. Its last coefficient is $c_{64} = -2816 \neq 0$, so no further tail satisfies anything shorter; any order-64 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316172 is "Number of nX4 0..1 arrays with every element unequal to 1, 2, 4, 5, 6 or 7 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$2, 10, 34, 72, 331, 1425, 5297, 18573, \ \dots$$

The entry states:

> Empirical recurrence of order 64 (see link above)

As of the "Last modified" line on the live entry (Jun 25 2018 17:37:57, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 70$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 70$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 64: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 64.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 140$ exact terms from $n = 3$ on, it returns order 64 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{64} c_i \, a(n-i),$$

c_1	5	c_{17}	2140147	c_{33}	−287858663	c_{49}	62443707
c_2	0	c_{18}	12024570	c_{34}	−286706259	c_{50}	−49966403
c_3	49	c_{19}	3705372	c_{35}	−76834888	c_{51}	−36822663
c_4	−325	c_{20}	−8434960	c_{36}	439510411	c_{52}	10252088
c_5	14	c_{21}	−34444257	c_{37}	80841261	c_{53}	16512965
c_6	−666	c_{22}	21188328	c_{38}	−111528385	c_{54}	−556800
c_7	8470	c_{23}	12728528	c_{39}	−16244875	c_{55}	−8268030
c_8	−900	c_{24}	41363666	c_{40}	276078389	c_{56}	782633
c_9	−3402	c_{25}	−83763076	c_{41}	−178035975	c_{57}	1884224
c_{10}	−115249	c_{26}	26554129	c_{42}	−449982344	c_{58}	858681
c_{11}	22207	c_{27}	−4503948	c_{43}	54121514	c_{59}	−331686
c_{12}	185340	c_{28}	96907045	c_{44}	374768162	c_{60}	−299880
c_{13}	874735	c_{29}	−149102810	c_{45}	97010767	c_{61}	8034
c_{14}	−290012	c_{30}	37787848	c_{46}	−225525634	c_{62}	12784
c_{15}	−2139187	c_{31}	76078710	c_{47}	−88553457	c_{63}	12592
c_{16}	−3444332	c_{32}	243025053	c_{48}	119209279	c_{64}	−2816

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 67$, and it is the recurrence OEIS A316172 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{64} - c_1 x^{64-1} - \cdots - c_{64}$. Its constant term is $-c_{64} = 2816$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 64 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 64, so $\deg q = 64$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 64, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 64, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -2816.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-64 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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